7 Fig. 7.1 shows a small mass attached to the end of a fixed spring.

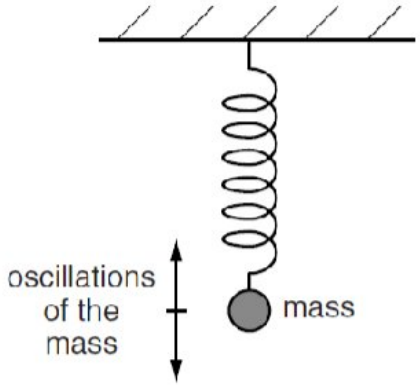


Fig. 7.1 (not to scale)

The mass is given a small vertical displacement and then released. It oscillates with simple harmonic motion about its equilibrium position.

(a) Define the term *simple harmonic motion*.

.....

.....

.....

..... [2]

(b) The graph in Fig. 7.2 shows how the displacement x varies with time t for the motion of the mass on the spring.

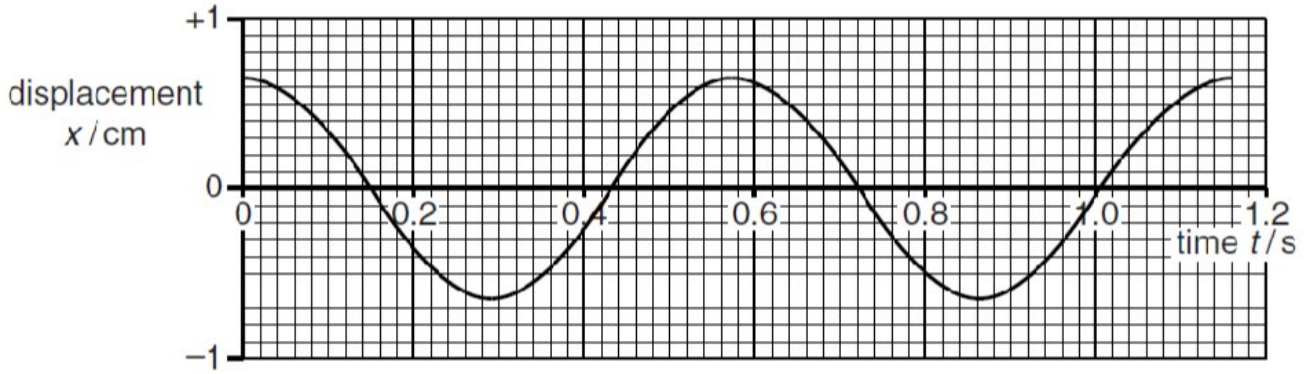


Fig. 7.2

(i) Using data from the graph in Fig. 7.2, determine the velocity of the mass at time $t = 0.66$ s.

velocity = ... cm s^{-1} [3]

(ii) Fig. 7.3 shows the spring and the position of the mass at instant $t = 0.94$ s.

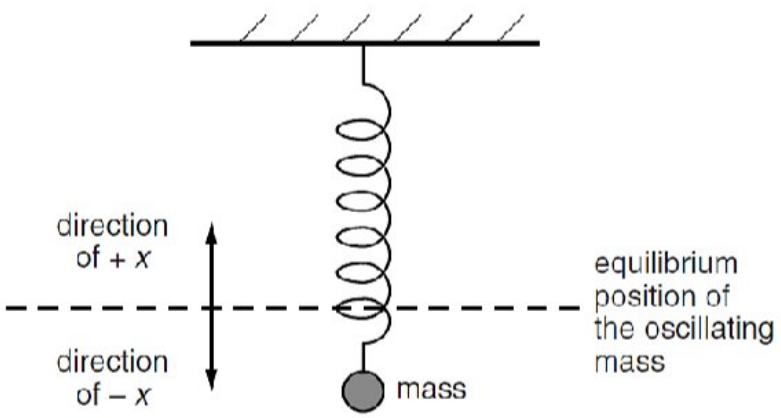


Fig. 7.3

On Fig. 7.3, add arrows labelled *a* and *v* to show the directions of the acceleration and the velocity, respectively, at this instant in time. [2]

(c) Fig. 7.4 shows a uniform rod floating vertically in water of density ρ. The rod has a mass M and a uniform cross-sectional area A.

When the rod is in equilibrium, its base is at a depth h below the water surface.

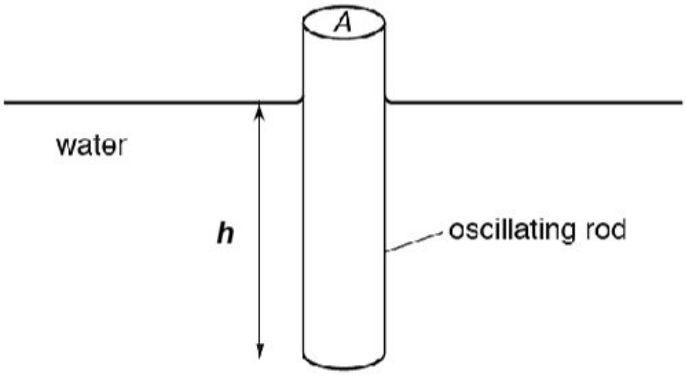


Fig. 7.4

The rod is displaced vertically and released. The acceleration a of the resulting oscillations is given by

$$a = -\left(\frac{\rho A g}{M}\right) x$$

where x is the displacement from the equilibrium depth h and g is the acceleration due to free fall.

(i) Show that the motion of the rod is simple harmonic.

[2]

(ii) Hence, derive an expression for the period T of the oscillations, in terms of ρ, M, A and g.

[2]

(iii) The rod is given a small vertical displacement and then released. It oscillates, but damping soon causes it to come to rest after a few oscillations in the water.

State why the damping causes the oscillations to stop.

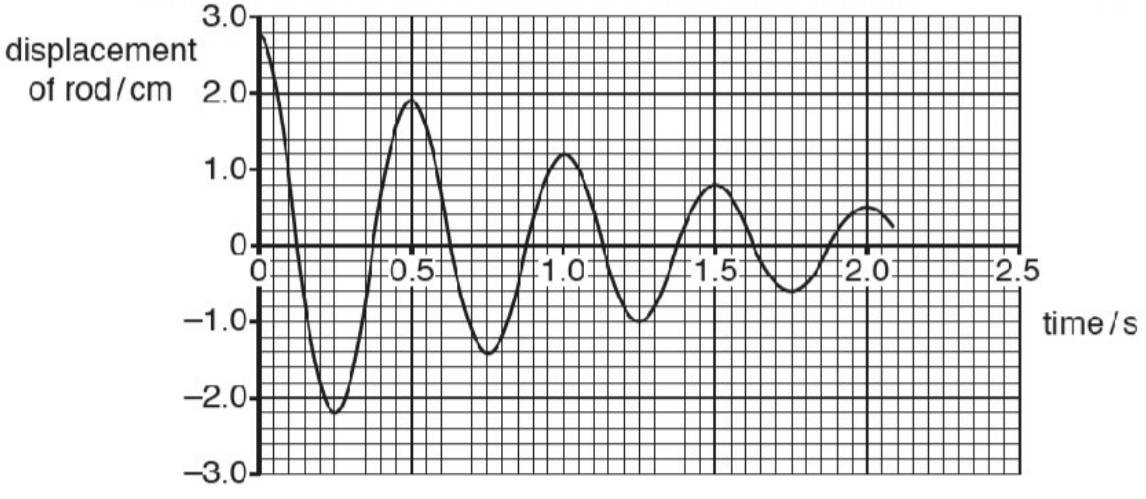
.....

.....

.....

.....[1]

(iv) Information from Fig. 7.5 can be used to verify that the damping on the rod is exponential.



1. Use Fig. 7.5 to complete the second column in the table of Fig. 7.6 to record the displacement at the times stated. [1]

2. Use the displacement data and the blank column in Fig. 7.6 to show that the rod exhibits light damping, where its amplitude decays exponentially with time. [1]

3. Use the grid in Fig. 7.7 to draw a graph that illustrates your answer. [2]

time / s	displacement / cm	
0.0		
0.5		
1.0		
1.5		
2.0		

Fig. 7.6

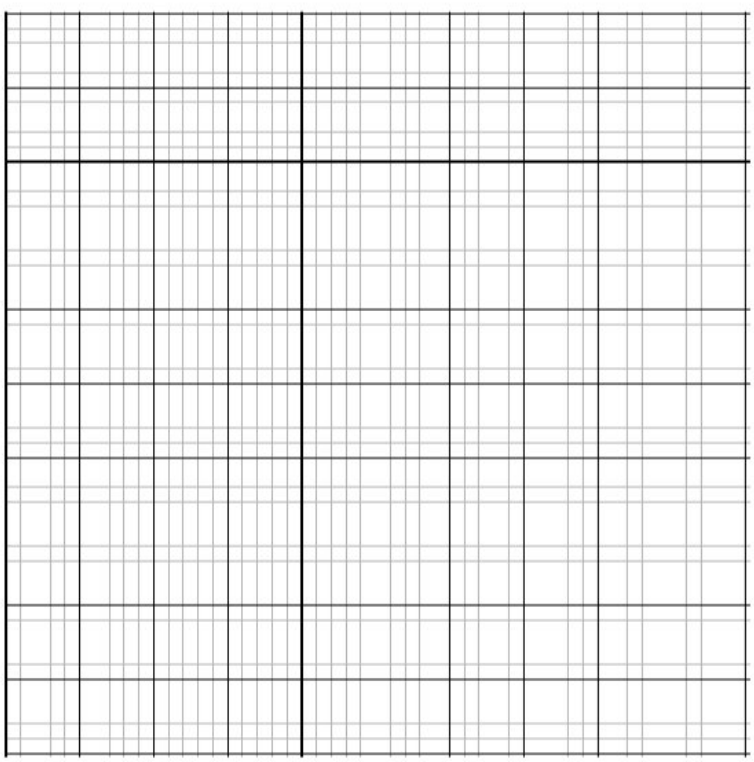


Fig. 7.7

(d) Earthquake waves can produce forced oscillations in tall buildings. The buildings may collapse because resonance occurs.

Explain what is meant by the term *resonance* in this situation.

.....

.....

.....

..... [2]

(e) Fig. 7.8 shows a tall building designed to prevent collapse during an earthquake. The building has a solid base which rests on a rubber bearing.

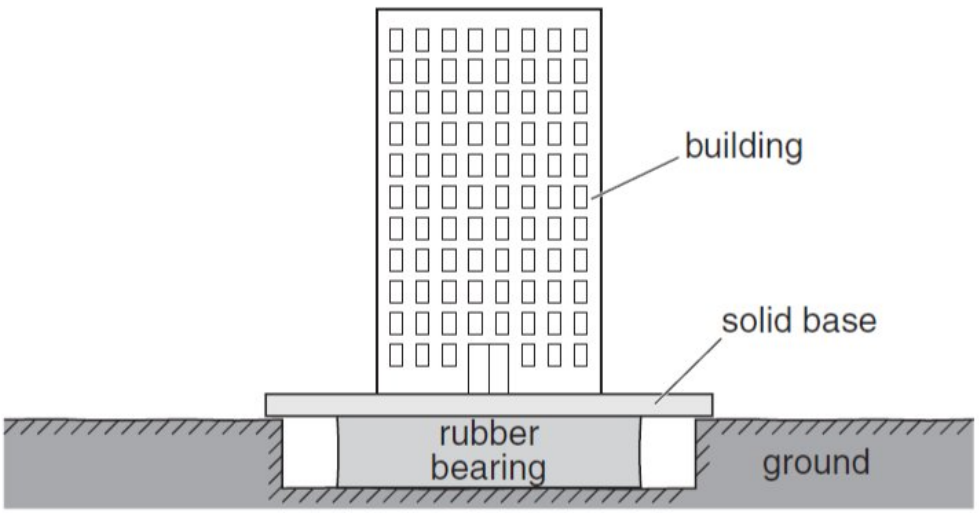


Fig. 7.8

Suggest why this design helps to prevent the building from collapse.

.....

.....

.....

..... [2]

[Total: 20]